



Lecture 0b

Math Foundations Team



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Vector



$$\underline{\vec{a} + \vec{b}}$$



- ▶ A point in $\mathbb{R}^n$ is represented by a n tuple i.e. $(x_1, \dots, x_n)$ can be considered as a column vector $[x_1, \dots, x_n]^T$ or a row vector $[x_1, \dots, x_n]$.
- ▶ We can define addition:
$$\mathbf{x} + \mathbf{y} = (x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$
- ▶ Multiplication by scalars:
$$\lambda \mathbf{x} = \lambda(x_1, x_2, \dots, x_n) = (\lambda x_1, \lambda x_2, \dots, \lambda x_n), \forall \lambda \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n.$$

Linear Combinations



$$\bar{x}, \bar{y}$$

$$\bar{x} + \bar{y}$$

$$c_1, c_2, \dots, c_m \in \mathbb{R}$$

$$c_1 v_1 + c_2 v_2 + \dots + c_m v_m$$

Let $v_1, \dots, v_m \in \mathbb{R}^n$ be m vectors, then a linear combination of these vectors is an expression

$$c_1 v_1 + \dots + c_m v_m = \sum_{i=1}^m c_i v_i = v$$

where c_i s are any scalars.

Example: $2[1, 3]^T + 3[1, 0]^T = [5, 6]^T$

$$c_1 v_1 + c_2 v_2 + \dots + c_m v_m$$

$$= \begin{bmatrix} v_1 & v_2 & \dots & v_m \end{bmatrix}_{n \times m} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}_{m \times 1} = v_{n \times 1} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



- * Consider

$$c_1 \mathbf{v}_1 + \cdots + c_m \mathbf{v}_m = \mathbf{0}$$

where $c_i \in \mathbb{R}$ and $\mathbf{v}_i \in \mathbb{R}^n$, $i = 1, \dots, n$.

- * Clearly if $c_i = 0$, $\forall i = 1, \dots, n$, then the above equation will hold.
Now, if no other c_i s satisfies the above equation, we say that $\mathbf{v}_i$ s are linearly independent.
- * If there exists c_i s that satisfies the above equation, and not all c_i s are zero, then we say that $\mathbf{v}_i$ s are linearly dependent.

Suppose $\mathbf{v}_i$ are linearly dependent $\Rightarrow$ There exists $j \in \{1, \dots, n\}$ such that

$$c_1 \mathbf{v}_1 + \dots + c_m \mathbf{v}_m = \mathbf{0} \text{ and } c_j \neq 0.$$

$$\Rightarrow \mathbf{v}_j = \sum_{i=1, i \neq j}^n \frac{-c_i}{c_j} \mathbf{v}_i$$

That means if $\mathbf{v}_i$ s are linearly dependent then we can express at least one of the vectors as linear combination of the remaining vectors.

Ex.1 Consider $[1, 3]^T, [1, 0]^T, [5, 6]^T$. Then

$$2[1, 3]^T + 3[1, 0]^T = [5, 6]^T$$

Therefore, $[1, 3]^T, [1, 0]^T, [5, 6]^T$ form a linearly dependent set.

Ex.2 Consider $[1, 3]^T, [1, 0]^T$. Let

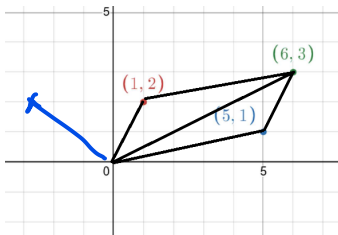
$$\begin{aligned} a[1, 3]^T + b[1, 0]^T &= [0, 0]^T \\ \Rightarrow [a + b, 3a]^T &= [0, 0]^T \\ \Rightarrow \underline{a = 0, b = 0} \end{aligned}$$

$$\begin{aligned} &\begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix} \\ &R_2 - 3R_1 \\ &\begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix} \end{aligned}$$

Therefore, $[1, 3]^T, [1, 0]^T$ form a linearly independent set.

Ex.3 Consider $[1, 2]^T, [5, 1]^T, [6, 3]^T$ Now,

$$[1, 2]^T + [5, 1]^T = [6, 3]^T$$



L.D

They form a linearly dependent set.



- * Consider p vectors such that each of these vectors have n components. then these vectors are linearly independent if the matrix form with these vectors as row vectors has rank p .
- * They are linearly dependent if $\text{rank} < p$.
- * The rank r of a matrix $\mathbf{A}$ is also equals the maximum number of linearly independent column vectors of $\mathbf{A}$
Therefore, $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^T) = r$.
- * Consider p vectors each having n components. If $n < p$ then these vectors are linearly dependent.

To check for linear independence



- ▶ A practical way of checking whether a bunch of vectors are linearly independent is to fill out the columns of a matrix with the given vectors and get the row-echelon form
- ▶ Pivot columns or Pivot rows are the columns or rows containing pivot elements.
- ▶ The pivot columns indicate all the vectors that are linearly independent, and they are all on the left.
- ▶ The non-pivot columns can be expressed as a linear combination of the pivot columns

$[v_1 \dots$

Example



- ▶ Let us start with the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- ▶ We get the following row-echelon form by elementary row operations

$$\begin{array}{ccc} p & np & p \\ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -2 \end{bmatrix} \end{array}$$

- ▶ The pivot columns are the first and the third column and we see that the second column which is a non-pivot column can be expressed as a linear combination of the pivot columns to its left, which is just twice the first column.

Another example



Consider the vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}$

To check for linear independence we set up the equation with λ s as follows: $\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \lambda_3 \mathbf{x}_3 = \mathbf{0}$.

As before we put the given vectors into a matrix to get:

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & -2 \\ -3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$AX = 0$ is const
 $0 = 0$
 $AX = 0$
unique
L.I.
const

. Now, we can look into REF/RREF to find whether these vectors are linearly independent or not.

Dot Product

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2$$

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$



$$\cos \theta =$$



$$\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = 0$$

$$\theta = 90^\circ \quad \theta = \frac{\pi}{2}$$

If $\mathbf{a}$ and $\mathbf{b}$ are vectors in $\mathbb{R}^n$, then $\mathbf{a}^T \mathbf{b}$ is called the dot product of $\mathbf{a}$ and $\mathbf{b}$.

Notation: $\langle \mathbf{a}, \mathbf{b} \rangle$ or $\mathbf{a} \cdot \mathbf{b}$.

L

$$\Rightarrow \mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = [a_1, \dots, a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^n a_i b_i$$

For example: $\mathbf{a} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, then

$$\mathbf{a} \cdot \mathbf{b} = (1)(-1) + (0)(2) + (2)(1) = 1.$$

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow \mathbf{a} \perp \mathbf{b}$$

Properties of Dot Product



$$a \cdot b = \langle a, b \rangle$$

$$a \cdot b = b \cdot a \quad a \cdot a \geq 0 \quad \text{if } = 0 \Rightarrow a = 0 \quad \text{ka} \cdot b$$
$$(a+b) \cdot c$$

- ▶ $\langle k\mathbf{u} + l\mathbf{v}, \mathbf{w} \rangle = k\langle \mathbf{u}, \mathbf{w} \rangle + l\langle \mathbf{v}, \mathbf{w} \rangle, \forall k, l \in \mathbb{R}, \forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$
(linearity) ✓
- ▶ $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ (symmetry) ✓
- ▶ $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0, \forall \mathbf{u} \in \mathbb{R}^n$
 $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$ (positive definite) ✓



$$\|a\| = \sqrt{a \cdot a} = \sqrt{a_1^2 + \dots + a_n^2}$$

The length or norm of a vector $\mathbf{a} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} \geq 0.$$

$$|s| \leq r \\ -r \leq s \leq r$$

Properties

- ▶ $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$ Cauchy Schwarz inequality.
- ▶ $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$ Triangle inequality.

$$-\|\mathbf{a}\| \cdot \|\mathbf{b}\| \leq \langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\| \cdot \|\mathbf{b}\|$$

$$-1 \leq \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|} \leq 1$$

Angle between 2 vectors and orthogonality



Now we have for any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, we have

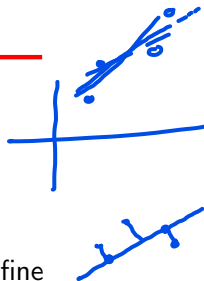
$$-1 \leq \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|} \leq 1$$

Let α be the angle between $\mathbf{a}, \mathbf{b}$. Then, we can define

$$\alpha = \cos^{-1} \left(\frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|} \right)$$

Clearly, $\langle \mathbf{a}, \mathbf{b} \rangle = 0 \Rightarrow \alpha = \frac{\pi}{2}$. Then we say $\mathbf{a}, \mathbf{b}$ are orthogonal to each other.

For example: Consider $[2, 2]^T, [2, -2]^T$. Clearly $[2, 2][2, -2]^T = 0$ and so they are orthogonal to each other.



Projection

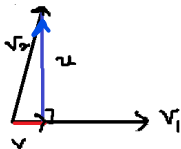


$$\mathbf{v}_2 = \mathbf{v} + \mathbf{u}$$

$$\mathbf{v} \perp \mathbf{u} \Rightarrow \mathbf{u} \perp \mathbf{v}_1$$

$$\Rightarrow \mathbf{u} = \mathbf{v}_2 - \mathbf{v}$$

dirⁿ of $\mathbf{v}$
is same as $\mathbf{v}_1$



$$\mathbf{v} = \lambda \cdot \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$$

The vector $\mathbf{v}$ is called as the projection of $\mathbf{v}_2$ onto the vector $\mathbf{v}_1$.

Then $\mathbf{u} = \mathbf{v}_2 - \mathbf{v}$ is orthogonal to $\mathbf{v}_1$ and $\mathbf{v} = \lambda \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$ for some scalar λ .

$$\mathbf{u}^T \mathbf{v}_1 = 0$$

$$\mathbf{v}^T \mathbf{v}_1 = \lambda \frac{\mathbf{v}_1^T \mathbf{v}_1}{\|\mathbf{v}_1\|}$$

$$\Rightarrow (\mathbf{v}_2 - \mathbf{v})^T \mathbf{v}_1 = \mathbf{v}_2^T \mathbf{v}_1 - \mathbf{v}^T \mathbf{v}_1 = 0$$

$$\Rightarrow \mathbf{v}_2^T \mathbf{v}_1 = \mathbf{v}^T \mathbf{v}_1 = \lambda \|\mathbf{v}_1\| \Rightarrow \lambda = \frac{\mathbf{v}_2^T \mathbf{v}_1}{\|\mathbf{v}_1\|}$$

$$\Rightarrow \mathbf{v} = \frac{\mathbf{v}_2^T \mathbf{v}_1}{\|\mathbf{v}_1\|} \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{\mathbf{v}_2^T \mathbf{v}_1}{\mathbf{v}_1^T \mathbf{v}_1} \mathbf{v}_1$$

$$\mathbf{v} = \frac{\mathbf{v}_2^T \mathbf{v}_1}{\mathbf{v}_1^T \mathbf{v}_1} \mathbf{v}_1$$

Basic Terminologies in Probability

$\{1, 2, 3, 4, 5, 6\}$



- ▶ Sample Space: The set of all possible outcomes of a random experiment is known as the sample space of the experiment and is denoted by Ω .
 $\{1, 2, 3, 4, 5, 6\}$
- ▶ Events: Any subset E of sample space Ω of a random experiment is known as event.
 $\{1, 2, 3\}$ $\{4, 5, 6\}$ $\{3, 4\}$
- ▶ Algebra of Events: Union and intersection of finitely many events is an event. Complement of an event is an event.
- ▶ Mutually exclusive events: The collection of events $\{E_1, E_2, E_3, \dots\}$ is said to be mutually exclusive, if $E_i \cap E_j = \phi$, for all $i \neq j$.
- ▶ Exhaustive Events: The collection of events $\{A_1, A_2, \dots, A_n\}$ is said to be exhaustive if $\cup_{i=1}^n A_i = \Omega$, where Ω is the sample space of the experiment.

Definition of Probability

$\mathcal{B}$: set of subsets of Ω
 $\mathcal{B} = \{\Omega, \emptyset\}$ $\mathcal{B}$ = Power set of Ω



Let Ω be any sample space and $\mathcal{B}$ be an algebra of events. A set function $P : \mathcal{B} \rightarrow \mathbb{R}$ is said to be a probability function if it satisfies the following three axioms:

► Axiom 1: $P(E) \geq 0$ for all $E \in \mathcal{B}$.

► Axiom 2: $P(\Omega) = 1$

► Axiom 3: For any sequence of mutually exclusive events $E_1, E_2, \dots$,

$\mathcal{B} = \{\Omega, \emptyset, A, A^c\}$
 where $A \subseteq \Omega$

$$P(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$$

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\mathcal{B} = \{\underline{\Omega}, \emptyset, \{1, 3, 5\}, \{2, 4, 6\}\}$$

Random Variable

$$X : \Omega \longrightarrow \mathbb{R}$$
$$X(\omega) = r$$



- ▶ Random variable is a real-valued function from a sample space Ω . We use uppercase letters to denote a random variable and lowercase letter to denote the numerical values observed by random variable (rv).
- ▶ A random variable is discrete if it can assume at most a finite or countably infinite numbers of possible values.
- ▶ Example: Number of heads when a coin is tossed thrice.
- ▶ A random variable is continuous if it assumes any value in some interval or intervals of real numbers and the
- ▶ Example: The lifetime of a transistor probability that it assumes any specific value is 0.



- ▶ The probability mass function (pmf) of a discrete random variable is defined for every x by

$$p(x) = P(X = x) = P(\{s \in \Omega : X(s) = x\}) \geq 0$$
$$\sum_x p(x) = 1$$

where Ω is the sample space.

- ▶ Let X be continuous random variable. A real valued integrable function f on $\mathbb{R}$ such that

$$f(x) \geq 0, \forall x \in \mathbb{R}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1,$$

$$P[a \leq X \leq b] = \int_a^b f(x) dx \text{ for } a, b \in \mathbb{R}$$

is called a density function for X .



- ▶ Measure of central tendency provides a very convenient way of describing a set of scores with a single number that describes the PERFORMANCE of the group. Mean is one such measure.
- ▶ The expectation or expected value of X , denoted by $E[X]$, is defined by

$$\mu = E[X] = \sum_x x \underline{p(x)} \text{ or } \int_{-\infty}^{\infty} x f(x) dx$$

- ▶ Empirically, based on observations, mean is computed as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \underline{\underline{\sum r_i \left(\frac{1}{n}\right)}}$$

where $x_i, i = 1, \dots, n$ are the n observations.



- ▶ Most common and most important measure of variability is the standard deviation.
- ▶ The variance of X is defined by

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 p(x) \text{ or } \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

- ▶ Empirically, based on observations, variance is computed as

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

- ▶ The standard deviation is defined as the squareroot of variance.



- ▶ In bivariate and multivariate case, the statistical technique used for studying the existence of relationship between all variables is through Covariance.
- ▶ If X and Y are two random variables, the covariance between X and Y is defined as

$$\text{cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Therefore, empirically, it is computed as

$$\text{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

where the ordered pair $(x_i, y_i) i = 1, \dots, n$ are the n observations.